COS 102: Problem Solving (Assignment)
Instructor: Anozie Onyezewe

Assignment Instructions (for All Groups)

Each group is required to complete the problem-solving steps for your given problem:

- Understand the problem: Clearly identify and document the inputs, outputs, and constraints (if any).
- Design a solution: Design a flowchart and pseudocode for your solution.
- Implement the solution: Write and test the code using Python (notebooks or VS Code).
- Evaluate and refine: Run test cases, debug any issues, and optimize your solution.

Provide a short report (2-3 pages) detailing your understanding of the problem, pseudocode, and flowchart of your solution, as well as your evaluation. Include the contribution of each group member at the bottom of your report. Mark absent group members as absent.

Group leaders are to submit your report (in PDF) and your code (.ipynb or .py) together in a zipped folder by the assigned deadline using this link: https://forms.gle/rnLMdGcoknzoZWBk8.

Deadline: Thursday, April 17, 2025, EOD.

Problem Assignments

Group 1: Perfect Square Checker

Problem Statement:

Given an integer N, determine if it is a perfect square.

Example: If N = 16, since 4 × 4 = 16, output = "Perfect Square". If N = 20, output = "Not a Perfect Square".

Group 2: Sum of Prime Numbers

Given an integer N, compute the sum of all prime numbers from 1 to N.

Example: Input: N = 10, Output: 17 (since the prime numbers ≤ 10 are 2, 3, 5, 7, and their sum is 17).

Group 3: Maximum Number Finder

Problem Statement:

Given a list of integers, find the maximum value in the list.

Example: For [3, 7, 2, 9, 5], output = 9.

Group 4: Vowel Counter in a String

Problem Statement:

Count the number of vowels (a, e, i, o, u) in a given string.

Example: For "Hello World", output = 3.

Group 5: String Reversal

Problem Statement:

Reverse the input string and display the reversed version.

Example: For "Data", output = "ataD".

Group 6: Prime Number Checker

Problem Statement:

Determine whether a given integer is a prime number.

Example: For N = 13, output = "Prime"; for N = 15, output = "Not Prime".

Group 7: Fibonacci Sequence Generator

Problem Statement:

Generate the first N numbers of the Fibonacci sequence.

Example: For N = 7, output = 0, 1, 1, 2, 3, 5, 8.

Group 8: Palindrome Checker

Problem Statement:

Check if a given string is a palindrome (reads the same forward and backward).

Example: For "radar", output = "Palindrome"; for "hello", output = "Not Palindrome".

Group 9: Bubble Sort Implementation

Problem Statement:

Sort a given list of numbers using the Bubble Sort algorithm.

Example: For [5, 3, 8, 4, 2], output = [2, 3, 4, 5, 8].

Group 10: Area and Perimeter of a Rectangle

Problem Statement:

Given the length and breadth of a rectangle, calculate both its area and perimeter.

Example: For length = 5 and breadth = 3, output: Area = 15, Perimeter = 16.

Group 11: Temperature Converter

Problem Statement:

Convert a temperature given in Celsius to Fahrenheit.

Formula: Fahrenheit = (Celsius × 9/5) + 32.

Example: For 25°C, output ≈ 77°F.

Group 12: Largest and Smallest in an Array

Problem Statement:

Given a list of numbers, find both the largest and smallest numbers.

Example: For [10, 3, 15, 7, 8], output: Largest = 15, Smallest = 3.

Group 13: Student Marks Analyzer

Problem Statement:

Given a list of student marks, calculate the average and determine the number of students passing (pass mark ≥ 50).

Example: For marks [45, 67, 80, 39, 90], output: Average ≈ 64.2; Pass Count = 3.

Group 14: Simple Calculator

Problem Statement:

Create a program that functions as a simple calculator performing addition, subtraction, multiplication, and division based on user input.

Example: Given inputs 8 and 4, it should correctly output 12, 4, 32, and 2 respectively for the four operations.

Group 15: Armstrong Number Checker

Problem Statement:

Given an integer, determine if it is an Armstrong number.

Definition: An Armstrong number is an integer such that the sum of the cubes of its digits equals the number itself.

Example: For 153, since $1^3 + 5^3 + 3^3 = 153$, output = "Armstrong Number"; for 123, output = "Not an Armstrong Number".

Group 16: Leap Year Checker

Problem Statement:

Determine if a given year is a leap year.

Criteria: A year is a leap year if it is divisible by 4, but not by 100 unless also divisible by 400.

Example: For 2000, output = "Leap Year"; for 1900, output = "Not a Leap Year".

Group 17: Word Frequency Counter

Problem Statement:

Count the frequency of each word in a given sentence.

Example: For "data science is fun and data is powerful", output might be: data: 2, science: 1, is: 2, fun: 1, and: 1, powerful: 1.

Group 18: GCD and LCM Calculator

Problem Statement:

Given two integers, calculate both their Greatest Common Divisor (GCD) and Least Common Multiple (LCM).

Example: For numbers 12 and 18, output: GCD = 6, LCM = 36.

Group 19: Prime Numbers Within a Range

Problem Statement:

Given two integers representing a range, find all prime numbers within that range.

Example: For the range 10 to 30, output = [11, 13, 17, 19, 23, 29].